

ACTIVITY: What is the “Mystery Group”?

$*$	A	B	C	D	E	F
A	B	A	D	C	F	E
B	A	B	C	D	E	F
C	F	C	B	E	D	A
D	E	D	A	F	C	B
E	D	E	F	A	B	C
F	C	F	E	B	A	D

Definition. An isomorphism ϕ from a group G to $\bar{G}$ is:

Question 1. What properties do you think isomorphic groups have?

To prove two groups are isomorphic:

Step 1:

Step 2:

Step 3:

Step 4:

Example 1. Prove that $M = \left\{ \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} : a \in \mathbb{Z} \right\}$ under matrix multiplication is isomorphic to $\mathbb{Z}$.

Theorem (Cayley's Theorem (1854)).

Theorem 6.2 Properties of Isomorphisms Acting on Elements.

Let $G, \bar{G}$ be groups with respective identities $e, \bar{e}$. Let $k, n \in \mathbb{Z}$ and $a, b \in G$. Then

1. $\phi(e) = \bar{e}$;
2. $\phi(a^n) = [\phi(a)]^n$;
3. $ab = ba$ iff $\phi(a)\phi(b) = \phi(b)\phi(a)$;
4. $G = \langle a \rangle$ iff $\bar{G} = \langle \phi(a) \rangle$;
5. $|a| = |\phi(a)|$; and
6. $|\{x \in G \mid x^k = b\}| = |\{x \in \bar{G} \mid x^k = \phi(b)\}|$.

Exercise 1. Prove #1.

Exercise 2. Prove #3.

Exercise 3. Prove #4.

Definition. An automorphism is:

Exercise 4. How many automorphisms can you find from Z_{12} onto itself? What are they?

Definition. Let G be a group and $a \in G$. The inner automorphism of G induced by a , denoted ϕ_a , is:

Exercise 5. Determine the inner automorphism of S_3 induced by (12) .

Theorem.

Theorem.

Additional Exercises

1. Prove that if G is a group, the mapping $\phi(g) = g^{-1}$ is an isomorphism if and only if G is Abelian.

2. Determine $\text{Aut}(\mathbb{Z})$.

3. For inner automorphism ϕ_g, ϕ_h , and ϕ_{gh} , prove that $\phi_g \phi_h = \phi_{gh}$.
4. Suppose $\phi \in \text{Aut}(\mathbb{Z}_n)$ and a is relatively prime to n . If $\phi(a) = b$, determine a formula for $\phi(x)$.
5. Let $\mathbb{R}^+$ be the positive real numbers under multiplication. Prove that $\phi(x) = \sqrt{x}$ is an automorphism of $\mathbb{R}^+$.

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6. Suppose G is a finite Abelian group and G has no element of order 2. Show the mapping $g \rightarrow g^2$ is an automorphism of G . Is this also true of groups with infinite order?
7. Suppose g and h induce the same inner automorphism of a group G . Prove that $h^{-1}g \in Z(G)$.
8. Prove that every automorphism of $\mathbb{R}^*$, the group of nonzero real numbers under multiplication, maps positive numbers to positive numbers and negative numbers to negative numbers.